Intrinsic Hyperspectral Image Recovery for UAV Strips Stitching

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***Abstract*— Due to various interferences from natural factors such as changes in the atmosphere and varying illumination conditions, a common issue arises in hyperspectral (HS) imaging, wherein the same material may exhibit different spectra. Unmanned aerial vehicle (UAV) HS imaging is also confronted with this challenge. Moreover, it is essential to merge high-resolution images captured by UAVs because of the limited width of each individual UAV image. This article discusses the aforementioned issues and presents a proficient intrinsic image decomposition (IID) technique for UAV strip stitching to achieve superior quality stitched intrinsic HS UAV images. Initially, an IID energy constraint is introduced for UAV strips to model the IID process of two HS images with a partially overlapping region. Subsequently, an anti-shadow interference mechanism is integrated to mitigate the impact of imaging perspective changes in the overlapping region. However, spectral drift may occur during IID operation in nonoverlapping areas since intrinsic properties recovered from nonoverlapping regions are relatively less reliable. To address this issue, a constraint for spectral drift correction is suggested to rectify the spectral bias existing between overlapping and nonoverlapping regions. The experimental results demonstrate that the proposed method effectively recovers intrinsic properties for UAV HS image strips.**

***Index Terms*— Hyperspectral (HS), intrinsic image decomposition (IID), stitching, unmanned aerial vehicle (UAV).**

I. Introduction

AT PRESENT, hyperspectral (HS) technology represents a primary development direction in the field of remote sensing [1]. HS imaging acquires data across a wide spectrum, usually covering the visible and near-infrared ranges, providing a greater number of spectral bands compared to conventional multispectral imaging. The wealth of spectral data available allows for the identification of materials or substances within a given scene. This capability aids in ground cover classification [2], [3], [4], target detection [5], [6], military surveillance [7], environmental monitoring [8], and characteristic analysis [9].

Currently, HS technology is undergoing a transition from satellite-based HS imaging to unmanned aerial vehicle (UAV) HS imaging [10]. On the one hand, satellite-based HS imaging offers a synoptic view of the ground surface with a long revisit period, facilitating extensive mapping and environmental monitoring on a large scale. However, it generally exhibits lower spatial resolution compared to high spatial resolution imaging using UAVs. On the other hand, UAV HS imaging provides a more detailed and localized view of the ground surface, with higher spatial resolution compared to satellite-based HS imaging. Therefore, UAV HS imaging is attracting growing interest in the field of remote sensing [11]. Nevertheless, the utilization of UAV HS imaging is limited by the flight path of the UAV, necessitating the stitched UAV HS image scene for subsequent more complex HS imaging tasks such as agriculture monitoring [12], mineral exploration [13], and geographical mapping [14].

In the context of UAV HS imaging, factors like environmental atmospheric conditions and illumination intensity tend to introduce spectral uncertainty into obtained HS images. The presence of uncertainty is manifested through fluctuations in the spectrum of a given material, which can pose a significant obstacle to HS image analysis tasks that depend on spectral characteristics [15], [16]. Intrinsic image decomposition (IID) is considered an effective approach to tackle this challenge. In this model, a natural image can be conceptualized as the product of two intrinsic components: reflectance and shading. Reflectance is associated with the properties of the corresponding material, while shading pertains to environmental interferences [17]. The main goal of IID is to separate the reflectance and shading components and extract reflectance component from a given image, which serves to enhance the analytical capabilities for ground features such as spectral unmixing and facilitates the precise calculation of reflectance derivatives such as normalized difference vegetation index (NDVI) [18], [19].

Since IID aims to separate the reflectance and shading components within a single image, the IID model initially encounters a typical ill-posed inversion problem due to its simultaneous inversion of two or more variables from a single variable. Therefore, although IID possesses the ability to extract spectral reflectance from HS images, the accuracy of extraction is still limited [20], [21]. Moreover, the intricate imaging factors inherent in remote sensing images can interfere with the decomposition model, consequently reducing its efficacy. In addition, HS images typically contain numerous spectral bands that need to be retrieved, further increasing the difficulty of reflectance inversion [22]. Therefore, employing a separate IID for each strip of

Manuscript received 17 February 2024; revised 15 April 2024 and 31 May 2024; accepted 25 June 2024. Date of publication 24 July 2024; date of current version 27 August 2024. This work was supported in part by the Distinguished Young Scholars of National Natural Science Foundation of China under Grant 62025107, in part by the Young Elite Scientists Sponsorship Program by the China Association for Science and Technology (CAST) under Grant 2022QNRC001, and in part by the Key Research and Development Program of Heilongjiang Province under Grant JD22A019. *(Corresponding author: Tianzhu Liu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3433024

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UAV imagery will yield inaccurate decomposition results and affect the subsequent utilization of the stitched image scene.

In fact, acquiring accurate reflectance directly from IID remains a challenging task. To enhance extraction accuracy, researchers have explored incorporating prior hypothesis information or available supplementary data into IID to establish more efficient models. The initial approach involved utilizing the hypothesis that the shading component is smooth to develop an enhanced IID model [23]. Moreover, the sparsity of reflectance in neighboring local areas is exploited to create new constraint items and improve the performance of IID [24], [25]. In addition, some studies even integrate both of these prior information into a unified model [26]. Practically, incorporating supplementary data provides another way for improving the inversion accuracy of IID model. Depth data, such as digital surface model (DSM), is utilized to promote the IID model [27], [28], [29], [30]. Point cloud data from light detection and ranging (LiDAR) provides one way for the IID model to remove reflectance shadows [31]. Images captured under varying illumination and light directions are also employed as inputs for IID [32], [33]. Furthermore, continuous-time video data is also utilized to extract and alter reflectance and shading components [34], [35]. Although these approaches enhance the capability of the IID model, they still face limitations when utilized for the purpose of stitching UAV strip images. On the one hand, IID models based on prior constraints typically perform decomposition from a single image, making it challenging to accurately recover the reflectance component and spectral inconsistency remains a problem when stitching intrinsic UAV strip images. On the other hand, since UAVs typically carry only one imaging sensor to obtain spectral data, IID methods utilizing data from other sensors as input are not applicable. To address these limitations, Gao et al. [36] proposed a bi-temporal IID (BIID) method, which aims to enhance the precision of reflectance extraction from bi-temporal HS images compared to a single HS image. However, it is constrained to HS images that share identical regions, which limits its applicability in scenarios involving the stitching of UAV image strips.

In light of these considerations, an HS IID model customized for UAV image strips is proposed to generate a stitched UAV reflectance component. Initially, an IID energy constraint is introduced for UAV image strips to represent the IID process of two partially overlapped HS images. Subsequently, an anti-shadow interference mechanism is incorporated to mitigate the impact of imaging perspective changes in overlapping region. Finally, a linear spectral drift correction IID model is utilized to rectify the spectral inconsistency between overlapping and nonoverlapping regions. Experimental results demonstrate that the proposed method effectively recovers the intrinsic properties of UAV HS image scene.

In summary, our main contribution is twofold as follows.

1) *IID Model for Partially Overlapped UAV Image Strips:* This model captures the intrinsic decomposition mechanism during the stitching process of UAV image scene and addresses the challenge posed by existing multi-temporal IID methods, which are unable to handle nonoverlapping regions, and resolves the spectral error propagation issue inherent in alternative stitching-IID two-step approaches.
2) *Anti-Shadow Interference Energy Coefficient:* The concept of an anti-shadow interference energy coefficient is introduced to characterize the light energy ratio of UAV image scene in overlapping region. This mechanism efficiently reduces shadow interference and enhances the performance of IID.

The rest of this article is organized as follows. In Section II, we introduce some related work. Section III presents our model in detail. Section IV shows the experimental results and related analysis. Section V draws conclusions for this article.

II. Related Work

In this section, some related work including HS IID work and UAV stitching work will be introduced in the following.

A. IID for HS Image (HIID)

For a single HS image, the relation between the intrinsic components of IID is as follows:

$$I = sR \tag{1}$$

where I is the input HS image, R is the reflectance component, and s is the shading component.

The basic HS IID model is formulated by two cost functions: one is data fidelity term, which is derived from formula (1) to ensure that the two intrinsic components adhere to the decomposition principle. The other is based on the prior hypothesis that the reflectance component tends to be similar in neighboring local area to enhance the capability of the IID model. The matrix form of the HIID model is expressed as

$$E(\boldsymbol{\rho}, \mathbf{B}) = \|\boldsymbol{\rho} + \mathbf{B} - \mathbf{O}\|_F^2 + \frac{1}{\sigma}\left\|\boldsymbol{\rho} - \boldsymbol{\rho}\mathbf{W}^T\right\|_F^2 \tag{2}$$

where $\boldsymbol{\rho}$ and $\mathbf{B}$ are reflectance and shading components in log domain, respectively, $\mathbf{O}$ is input HS image in log domain, and $\mathbf{W}$ is the matrix measuring the similarity relation of reflectance at local adjacent area.

For multitemporal HS images, when the interval between image acquisition times is short, there is essentially no change in ground cover, and the discrepancies between images are primarily attributed to data drift induced by natural factors. In this context, BIID is introduced based on the premise that multitemporal remote sensing images are expected to maintain constant reflectance at identical locations, with differences only existing in their shading components.

While BIID can extract more accurate reflectance component compared to HIID, it is limited to solving the IID problem within overlapping regions rather than the entire HS image scenes. Thus, the traditional BIID method is not suitable for addressing the IID problem associated with UAV strips.

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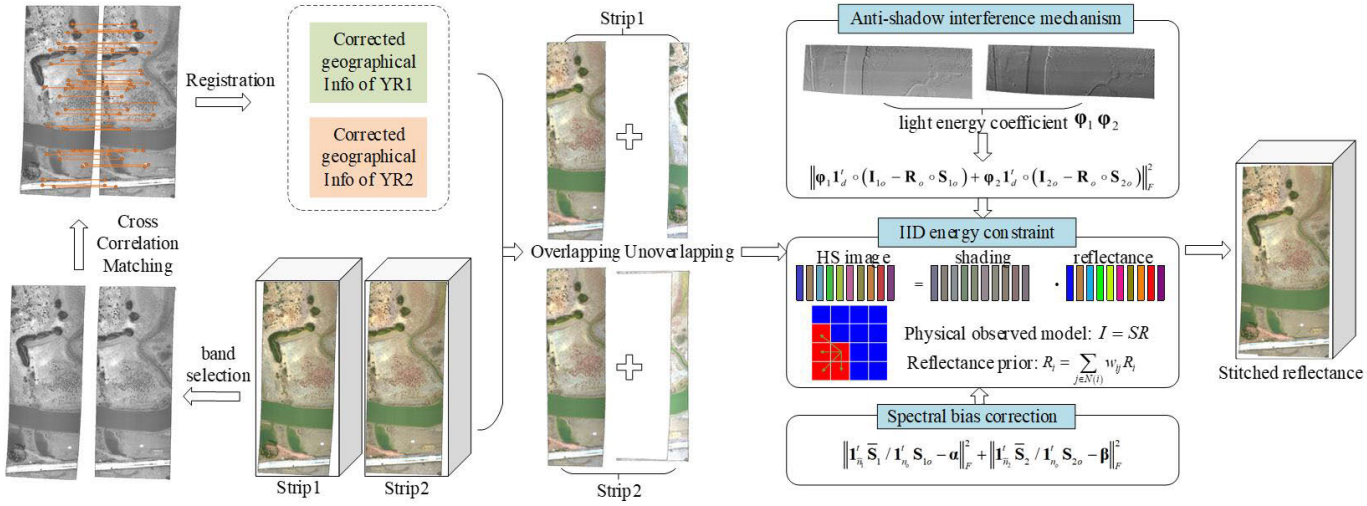


Fig. 1. Flowchart of the proposed method.

B. HS Images Stitching

UAV HS images, which contain both spatial and spectral information, have attracted considerable attention [37], [38], [39]. Given that UAV typically operates at low altitudes, capturing a wide viewing angle in a single image poses a significant challenge. Therefore, it is crucial to generate panoramic images to provide a comprehensive view of the environment over a large range [40], [41], [42], [43].

Recently, several HS image stitching methods tailored for UAV have been proposed. Fang et al. [44] introduced a UAV HS imaging system that is integrated with a stitching framework based on sparse bundle adjustment method. Mo et al. [45] designed an effective HS stitching method for UAV by employing deep feature matching and spectral correction based on intrinsic decomposition theory. Zhang et al. [46] proposed an automatic image stitching approach that utilizes mTopKRP algorithm and elastic warp to achieve high precision of feature matching in HS images. Liao and Li [47] introduced a single-perspective (SP)-based stitching method utilizing mesh-based warp and total energy function that emphasizes different characteristics of warp. In addition, Li et al. [48] employed a Bayesian model for outlier removal and implemented the thin plate spline method for analytical warping.

However, when confronted with the spectral inconsistency problem, these methods typically perform spectral correction on the target image to align its spectral range with that of the reference image rather than adhering to the actual reflectance spectrum values of the target image.

To address the issues encountered by the aforementioned works, this article provides an HS IID model for UAV image strips intended for stitching. This model is designed to eliminate interferences caused by the environment and to recover intrinsic reflectance components for the entire image scene.

III. Proposed Method

In this section, the proposed method will be described in detail. Fig. 1 illustrates the overall flowchart of this method, which consists of several main steps: First, the noise-free and high-quality spectral bands from two UAV image strips are utilized to perform cross-correlation matching. Second, overlapping and nonoverlapping areas are extracted from corrected image strips. Then, both anti-shadow interference mechanism and spectral bias correction prior are incorporated into the HS image intrinsic decomposition model. Finally, the energy function for UAV image strips is formulated and solved by iteratively finding the minimum Frobenius norm.

A. IID Energy Constraints for UAV Images

Two input HS images are given, denoted by $\mathbf{I}_1$ and $\mathbf{I}_2$, where $\mathbf{I}_i = [I(\lambda_1), I(\lambda_2), \ldots, I(\lambda_d)]^{\mathrm{T}}$ and d is the number of spectral bands of input HS image. Since the two input HS images are partially overlapped with each other, we extract the overlapping region of images $\mathbf{I}_1$ and $\mathbf{I}_2$ by overlapping masks. The overlapping part of $\mathbf{I}_1$ is denoted as $\mathbf{I}_{1o}$, and the nonoverlapping part of $\mathbf{I}_1$ is denoted as $\bar{\mathbf{I}}_1$, similarly, the overlapping part of $\mathbf{I}_2$ is $\mathbf{I}_{2o}$, and the nonoverlapping part of $\mathbf{I}_2$ is $\bar{\mathbf{I}}_2$.

First, we need to define the intrinsic properties decomposed from input HS image $\mathbf{I}_1$ as reflectance $\mathbf{R}_1$ and shading $\mathbf{S}_1$, respectively. Similarly, reflectance $\mathbf{R}_2$ and shading $\mathbf{S}_2$ are defined for HS image $\mathbf{I}_2$. Assume there is no land cover changes in overlapping image regions, and the ground material remains as it was. Since reflectance component directly corresponds to the original properties of the ground material, it is natural to find they share one reflectance component in overlapping region, which is denoted as $\mathbf{R}_o = \bigcap_{i=1}^{2} \mathbf{R}_i$, and the difference only exists in shading components $\mathbf{S}_{1o}$ and $\mathbf{S}_{2o}$. As for nonoverlapping region in image $\mathbf{I}_1$, its reflectance and shading components are denoted as $\bar{\mathbf{R}}_1$ and $\bar{\mathbf{S}}_1$, respectively. Similarly, reflectance $\bar{\mathbf{R}}_2$ and shading $\bar{\mathbf{S}}_2$ are denoted for nonoverlapping region in image $\mathbf{I}_2$.

For UAV HS images, illumination and shadow conditions typically remain consistent within local neighborhoods. Therefore, variations in image intensity within these regions primarily arise from spectral differences. In fact, pixels with

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similar intensity values tend to exhibit similar reflectance values. Thus, the reflectance value of a given pixel can be effectively represented through the weighted summation of its neighboring pixel values [49]. Furthermore, it is crucial to minimize the error between the product of common reflectance and shading with the original HS image for each temporal HS image. Building upon the aforementioned concepts, an IID energy constraint function is introduced, which aims to derive intrinsic reflectance from the input UAV HS images. The energy constraint function for UAV images is expressed as follows:

$$E = \frac{\sigma}{2}\left\|\left[\bar{\mathbf{R}}_1, \mathbf{R}_o, \bar{\mathbf{R}}_2\right] - \left[\mathbf{W}_1\bar{\mathbf{R}}_1, \mathbf{W}_o\mathbf{R}_o, \mathbf{W}_2\bar{\mathbf{R}}_2\right]\right\|_F^2 + \left\|\left[\bar{\mathbf{I}}_1, \mathbf{I}_{1o}, \mathbf{I}_{2o}, \bar{\mathbf{I}}_2\right] - \left[\bar{\mathbf{R}}_1, \mathbf{R}_o, \mathbf{R}_o, \bar{\mathbf{R}}_2\right] \circ \left[\bar{\mathbf{S}}_1, \mathbf{S}_{1o}, \mathbf{S}_{2o}, \bar{\mathbf{S}}_2\right]\right\|_F^2 \tag{3}$$

where notation $\circ$ denotes element-wise multiplication operator. $\mathbf{W}_1$, $\mathbf{W}_2$, and $\mathbf{W}_o$ stand for the weighted summation coefficient matrixes of reflectance $\bar{\mathbf{R}}_1$, $\bar{\mathbf{R}}_2$, and $\mathbf{R}_o$, respectively. For each element w_{ij} of coefficient matrix $\mathbf{W}_1$ and $\mathbf{W}_2$, the more similar the spectrum of pixels i and j, the closer its value is to 1, and its expression is as follows:

$$w_{ij} = \exp\left\{-\left[\left(\mathbf{V}_i - \mathbf{V}_j\right)^2/\sigma_{i\mathbf{V}}^2 + \mathrm{A}\left(\mathbf{I}_i, \mathbf{I}_j\right)^2/\sigma_{i\mathrm{A}}^2\right]\right\}$$
$$\mathrm{A}\left(\mathbf{I}_i, \mathbf{I}_j\right) = \arccos\left[\left(\mathbf{I}_i^t\mathbf{I}_j\right)/\left(\sqrt{\mathbf{I}_i^t\mathbf{I}_i}\sqrt{\mathbf{I}_j^t\mathbf{I}_j}\right)\right] \tag{4}$$

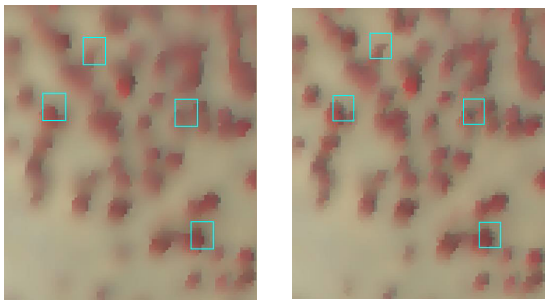
where $\mathbf{V}_i$ and $\mathbf{V}_j$ represent intensity value of pixel i and j, respectively, $\sigma_{i\mathbf{V}}^2$ and $\sigma_{i\mathrm{A}}^2$ represent variances of pixel intensities and spectral angle in local window, respectively, and the window size is usually set as 3 by 3.

B. Anti-Shadow Interference Mechanism

However, remote sensing images, especially high-resolution UAV images, often suffer from shadow interference. The reflectance extracted by IID is usually dim due to the low light energy, and this interference still exists even in the BIID model. To mitigate shadow interference, it is essential to reduce the reference coefficient of the overlapping region affected by shadows in the IID model for UAV images. Therefore, we introduce a light energy coefficient in the term of the IID energy constraint in overlapping region. The revised model is written as follows:

$$\begin{aligned} E &= \frac{\sigma}{2}\left\|\left[\bar{\mathbf{R}}_1, \mathbf{R}_o, \bar{\mathbf{R}}_2\right] - \left[\mathbf{W}_1\bar{\mathbf{R}}_1, \mathbf{W}_o\mathbf{R}_o, \mathbf{W}_2\bar{\mathbf{R}}_2\right]\right\|_F^2 \\ &+ \left\|\left[\bar{\mathbf{I}}_1, \bar{\mathbf{I}}_2\right] - \left[\bar{\mathbf{R}}_1, \bar{\mathbf{R}}_2\right] \circ \left[\bar{\mathbf{S}}_1, \bar{\mathbf{S}}_2\right]\right\|_F^2 \\ &+ \left\|\boldsymbol{\varphi}_1\mathbf{1}_d^t \circ \left(\mathbf{I}_{1o} - \mathbf{R}_o \circ \mathbf{S}_{1o}\right) + \boldsymbol{\varphi}_2\mathbf{1}_d^t \circ \left(\mathbf{I}_{2o} - \mathbf{R}_o \circ \mathbf{S}_{2o}\right)\right\|_F^2 \end{aligned} \tag{5}$$

where $\mathbf{1}_d^t$ denotes a 1 by d vector with all 1 s. $\boldsymbol{\varphi}_1$ and $\boldsymbol{\varphi}_2$ represent the light energy coefficient for overlapped HS images



(a) (b)

Fig. 2. Reflectance obtained by IID model (a) with anti-shadow interference constraint and (b) without anti-shadow interference constraint.

$\mathbf{I}_{1o}$ and $\mathbf{I}_{2o}$, respectively, and they are calculated by

$$\boldsymbol{\varphi}_1 = \sum_{k=1}^{d}(\mathbf{I}_{1o})_k / \sum_{k=1}^{d}(\mathbf{I}_{1o} + \mathbf{I}_{2o})_k$$
$$\boldsymbol{\varphi}_2 = \sum_{k=1}^{d}(\mathbf{I}_{2o})_k / \sum_{k=1}^{d}(\mathbf{I}_{1o} + \mathbf{I}_{2o})_k. \tag{6}$$

Furthermore, the shadow interference is also taken into account during the generation process of the weighted summation coefficient matrix for overlapping region. We modify the representation of the spectrum by incorporating a light energy coefficient for each pixel in the overlapping region, and each element of weighted summation coefficient matrix $\mathbf{W}_o$ can be written as

$$w_{ij}^o = \exp\left\{-\left[\left(\mathbf{V}_i^o - \mathbf{V}_j^o\right)^2/\sigma_{i\mathbf{V}}^2 + \mathrm{A}\left(\mathbf{I}_i^o, \mathbf{I}_i^o\right)^2/\sigma_{i\mathrm{A}}^2\right]\right\}$$
$$\mathbf{I}_i^o = \boldsymbol{\varphi}_i^1\mathbf{I}_i^{1o} + \boldsymbol{\varphi}_i^2\mathbf{I}_i^{2o}$$
$$\mathbf{I}_j^o = \boldsymbol{\varphi}_j^1\mathbf{I}_j^{1o} + \boldsymbol{\varphi}_j^2\mathbf{I}_j^{2o} \tag{7}$$

where $\mathbf{V}_i^o$ and $\mathbf{V}_j^o$ represent intensity value of $\mathbf{I}_i^o$ and $\mathbf{I}_j^o$, respectively.

Fig. 2 shows a demonstration of the effects of anti-shadow interference constraint on reflectance.

C. Spectral Bias Correction

As we know, the more images are used in the IID model, the more prior information is provided, which enhances the accuracy of extracting the reflectance component in principle. Thus, in conditions without changes of ground surface, the performance of IID model for multitemporal HS images is superior to that for a single HS image.

Similarly, in IID model for UAV images with partially overlapping region, the reflectance component $\mathbf{R}_o$ decomposed from overlapping region demonstrates higher accuracy compared to $\bar{\mathbf{R}}_1$ and $\bar{\mathbf{R}}_2$ decomposed from nonoverlapping regions, which have less HS images available and less prior information obtained than reflectance in overlapping region do.

Therefore, in order to eliminate the spectral bias between them, the decomposition result of overlapping region is utilized as a benchmark to rectify that of nonoverlapping regions. Since shading component is dependent on the illumination and geometric features of ground surface. For both overlapping and nonoverlapping regions, their shading components exhibit

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identical illumination intensity but different surface geometric distributions. Thus, the ratio of shading components $\bar{\mathbf{S}}_1$ and $\mathbf{S}_{1o}$ in spectral dimension remains constant, and can be obtained by single HS image decomposition method. This principle also holds true for $\bar{\mathbf{S}}_2$ and $\mathbf{S}_{2o}$. The constraint on shading components is as follows:

$$G = \left\| \mathbf{1}_{\bar{n}_1}^t \bar{\mathbf{S}}_1 / \mathbf{1}_{n_o}^t \mathbf{S}_{1o} - \boldsymbol{\alpha} \right\|_F^2 + \left\| \mathbf{1}_{\bar{n}_2}^t \bar{\mathbf{S}}_2 / \mathbf{1}_{n_o}^t \mathbf{S}_{2o} - \boldsymbol{\beta} \right\|_F^2 \tag{8}$$

$$\boldsymbol{\alpha} = \mathbf{1}_{\bar{n}_1}^t \exp\left(\ln \bar{\mathbf{I}}_1 - \lambda \bar{\bar{\mathbf{I}}}_1 \mathbf{G}_1^{-1}\right) / \mathbf{1}_{n_o}^t \exp\left(\ln \mathbf{I}_{1o} - \lambda \bar{\bar{\mathbf{I}}}_{1o} \mathbf{G}_o^{-1}\right) \tag{9}$$

$$\boldsymbol{\beta} = \mathbf{1}_{\bar{n}_2}^t \exp\left(\ln \bar{\mathbf{I}}_2 - \lambda \bar{\bar{\mathbf{I}}}_2 \mathbf{G}_2^{-1}\right) / \mathbf{1}_{n_o}^t \exp\left(\ln \mathbf{I}_{2o} - \lambda \bar{\bar{\mathbf{I}}}_{2o} \mathbf{G}_o^{-1}\right) \tag{10}$$

where n_o, $\bar{n}_1$, and $\bar{n}_2$ are the number of overlapping and nonoverlapping pixels in $\mathbf{I}_1$ and $\mathbf{I}_2$, respectively. Notation / denotes the element-wise division operator. It is remarkable that shading components in overlapping region $\mathbf{S}_{1o}$ and $\mathbf{S}_{2o}$ are used as reference fixed matrixes to optimize shading components in nonoverlapping regions $\bar{\mathbf{S}}_1$ and $\bar{\mathbf{S}}_2$. $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are constant matrix of size 1 by d, which correspond to the ratio of shading components in spectral dimension of overlapping and nonoverlapping regions in $\mathbf{I}_1$ and $\mathbf{I}_2$, respectively. $\bar{\bar{\mathbf{I}}}_1$ and $\bar{\bar{\mathbf{I}}}_2$ represent the HS images after spectral averaging and spatial averaging processes.

D. Solution of IID Problem

Combined with the previously introduced spectral bias correction constraints, the IID model (5) for UAV images can be revised as follows:

$$U = E\left(\bar{\mathbf{R}}_1, \mathbf{R}_o, \bar{\mathbf{R}}_2, \bar{\mathbf{S}}_1, \bar{\mathbf{S}}_2, \mathbf{S}_{1o}, \mathbf{S}_{2o}\right) + \frac{\delta}{2} G\left(\bar{\mathbf{S}}_1, \bar{\mathbf{S}}_2\right) \tag{11}$$

$$\min_{\bar{\mathbf{R}}_1, \mathbf{R}_o, \bar{\mathbf{R}}_2} U\left(\bar{\mathbf{R}}_1, \mathbf{R}_o, \bar{\mathbf{R}}_2, \bar{\mathbf{S}}_1, \bar{\mathbf{S}}_2, \mathbf{S}_{1o}, \mathbf{S}_{2o}\right). \tag{12}$$

The energy optimization of (12) represents a typical quadratic equation involving seven variables. By calculating the first-order derivative of each variable in U, its minimum value can be determined and the first partial derivatives of U are as follows:

$$\partial U / \partial \bar{\mathbf{R}}_1 = \sigma \mathbf{G}_1 \bar{\mathbf{R}}_1 + \bar{\mathbf{R}}_1 - \bar{\mathbf{I}}_1 / \bar{\mathbf{S}}_1 \tag{13}$$

$$\partial U / \partial \bar{\mathbf{R}}_2 = \sigma \mathbf{G}_2 \bar{\mathbf{R}}_2 + \bar{\mathbf{R}}_2 - \bar{\mathbf{I}}_2 / \bar{\mathbf{S}}_2 \tag{14}$$

$$\partial U / \partial \mathbf{R}_o = \sigma \mathbf{G}_o \mathbf{R}_o + \mathbf{R}_o - \frac{\boldsymbol{\varphi}_1 \mathbf{1}_d^t \circ \mathbf{I}_{1o} + \boldsymbol{\varphi}_2 \mathbf{1}_d^t \circ \mathbf{I}_{2o}}{\boldsymbol{\varphi}_1 \mathbf{1}_d^t \circ \mathbf{S}_{1o} + \boldsymbol{\varphi}_2 \mathbf{1}_d^t \circ \mathbf{S}_{2o}} \tag{15}$$

$$\partial U / \partial \bar{\mathbf{S}}_1 = \bar{\mathbf{S}}_1 + \delta \mathbf{1}_{\bar{n}_1} \left[\mathbf{1}_{\bar{n}_1}^t \bar{\mathbf{S}}_1 - \boldsymbol{\alpha} \circ \left(\mathbf{1}_{n_o}^t \mathbf{S}_{1o}\right)\right] - \bar{\mathbf{I}}_1 / \bar{\mathbf{R}}_1 \tag{16}$$

$$\partial U / \partial \bar{\mathbf{S}}_2 = \bar{\mathbf{S}}_2 + \delta \mathbf{1}_{\bar{n}_2} \left[\mathbf{1}_{\bar{n}_2}^t \bar{\mathbf{S}}_2 - \boldsymbol{\beta} \circ \left(\mathbf{1}_{n_o}^t \mathbf{S}_{2o}\right)\right] - \bar{\mathbf{I}}_2 / \bar{\mathbf{R}}_2 \tag{17}$$

$$\partial U / \partial \mathbf{S}_{1o} = \mathbf{S}_{1o} - \frac{\boldsymbol{\varphi}_1 \mathbf{1}_d^t \circ \mathbf{I}_{1o} + \boldsymbol{\varphi}_2 \mathbf{1}_d^t \circ \mathbf{I}_{2o}}{\boldsymbol{\varphi}_1 \mathbf{1}_d^t \circ \mathbf{R}_o} + \frac{\boldsymbol{\varphi}_2 \mathbf{1}_d^t \circ \mathbf{S}_{2o}}{\boldsymbol{\varphi}_1 \mathbf{1}_d^t} \tag{18}$$

$$\partial U / \partial \mathbf{S}_{2o} = \mathbf{S}_{2o} - \frac{\boldsymbol{\varphi}_1 \mathbf{1}_d^t \circ \mathbf{I}_{1o} + \boldsymbol{\varphi}_2 \mathbf{1}_d^t \circ \mathbf{I}_{2o}}{\boldsymbol{\varphi}_2 \mathbf{1}_d^t \circ \mathbf{R}_o} + \frac{\boldsymbol{\varphi}_1 \mathbf{1}_d^t \circ \mathbf{S}_{1o}}{\boldsymbol{\varphi}_2 \mathbf{1}_d^t} \tag{19}$$

$$\mathbf{G} = \mathbf{I} + \mathbf{W}\mathbf{W}^T - \mathbf{W} - \mathbf{W}^T \tag{20}$$

where $\mathbf{G}_1$, $\mathbf{G}_2$, and $\mathbf{G}_o$ are square matrixes with size of $\bar{n}_1 \times \bar{n}_1$, $\bar{n}_2 \times \bar{n}_2$, and $n_o \times n_o$, respectively, and $\mathbf{I}$ denotes a unit matrix.

By finding the zero position of these first derivatives of (13)–(19), the optimal solution of IID model (12) can be obtained iteratively, and the iteration functions of unknown variables are presented below

$$\bar{\mathbf{R}}_1 = (\sigma \mathbf{G}_1 + \mathbf{I})^{-1} \left(\bar{\mathbf{I}}_1 / \bar{\mathbf{S}}_1\right) \tag{21}$$

$$\bar{\mathbf{R}}_2 = (\sigma \mathbf{G}_2 + \mathbf{I})^{-1} \left(\bar{\mathbf{I}}_2 / \bar{\mathbf{S}}_2\right) \tag{22}$$

$$\mathbf{R}_o = (\sigma \mathbf{G}_o + \mathbf{I})^{-1} \frac{\boldsymbol{\varphi}_1 \mathbf{1}_d^t \circ \mathbf{I}_{1o} + \boldsymbol{\varphi}_2 \mathbf{1}_d^t \circ \mathbf{I}_{2o}}{\boldsymbol{\varphi}_1 \mathbf{1}_d^t \circ \mathbf{S}_{1o} + \boldsymbol{\varphi}_2 \mathbf{1}_d^t \circ \mathbf{S}_{2o}} \tag{23}$$

$$\bar{\mathbf{S}}_1 = \left(\mathbf{I} + \delta \mathbf{1}_{\bar{n}_1} \mathbf{1}_{\bar{n}_1}^t\right)^{-1} \left(\bar{\mathbf{I}}_1 / \bar{\mathbf{R}}_1 + \delta \mathbf{1}_{\bar{n}_1} \left[\boldsymbol{\alpha} \circ \left(\mathbf{1}_{n_o}^t \mathbf{S}_{1o}\right)\right]\right) \tag{24}$$

$$\bar{\mathbf{S}}_2 = \left(\mathbf{I} + \delta \mathbf{1}_{\bar{n}_2} \mathbf{1}_{\bar{n}_2}^t\right)^{-1} \left(\bar{\mathbf{I}}_2 / \bar{\mathbf{R}}_2 + \delta \mathbf{1}_{\bar{n}_2} \left[\boldsymbol{\beta} \circ \left(\mathbf{1}_{n_o}^t \mathbf{S}_{2o}\right)\right]\right) \tag{25}$$

$$\mathbf{S}_{1o} = \frac{\boldsymbol{\varphi}_1 \mathbf{1}_d^t \circ \mathbf{I}_{1o} + \boldsymbol{\varphi}_2 \mathbf{1}_d^t \circ \mathbf{I}_{2o}}{\boldsymbol{\varphi}_1 \mathbf{1}_d^t \circ \mathbf{R}_o} - \frac{\boldsymbol{\varphi}_2 \mathbf{1}_d^t \circ \mathbf{S}_{2o}}{\boldsymbol{\varphi}_1 \mathbf{1}_d^t} \tag{26}$$

$$\mathbf{S}_{2o} = \frac{\boldsymbol{\varphi}_1 \mathbf{1}_d^t \circ \mathbf{I}_{1o} + \boldsymbol{\varphi}_2 \mathbf{1}_d^t \circ \mathbf{I}_{2o}}{\boldsymbol{\varphi}_2 \mathbf{1}_d^t \circ \mathbf{R}_o} - \frac{\boldsymbol{\varphi}_1 \mathbf{1}_d^t \circ \mathbf{S}_{1o}}{\boldsymbol{\varphi}_2 \mathbf{1}_d^t}. \tag{27}$$

To achieve quicker convergence toward the optimal solution, the initial value of reflectance component is initialized as the input HS image, shading component is initialized as matrix with all 1 s, iterative stopping parameter ε is set as 10^{-4}, σ is in the rage of 1 to 10, and δ is empirically set as 0.5. In this article, the maximum number of iterations is typically set to 100 to prevent excessive calculation time.

IV. Experimental Results and Analysis

In this section, experimental results on real UAV image data are presented to assess the effectiveness of the proposed IID method for UAV images that are partially overlapped.

A. Experimental Equipment and Data Description

An integrated data acquisition system with ideal weight, size, and power consumption is used in this research. The system is composed of a Headwall Nano-Hyperspec push-broom HS scanner and a position and orientation system (POS) [50], and both are equipped on UAV platform. With the assistance of this data acquisition system, two datasets including HIT Campus and Shankou Mangrove are successfully acquired. The detail of this system is shown in Fig. 3.

Trimble APX-15 is also integrated to provide attitude and position information for imaging sensor [51], while POSPAC is most useful during postprocessing of raw data. As for detailed parameters of HS scanner, it has a 7.4-μm pixel pitch and 783-g weight and captures ground surface images at a bit depth of 12 bits with a wavelength range from 400 to 1000 nm. And DJI Matrice 600 Pro is used as the UAV platform with an attitude stabilizer.

1) Yellow River Delta (YR): This UAV HS dataset was acquired by Hunan University and captured using a Headwall Nano HS imaging sensor over the YR delta in Shandong province of China in 2020. It contains 270 bands at a spatial resolution of 0.1-m ranging from 400 to 1000 nm. Seven strips were captured over several days, and two of them, which

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Fig. 3. Integrated data acquisition system. (a) DJI Matrice 600 Pro UAV platform. (b) Headwall Nano-Hyperspec and APX-15.

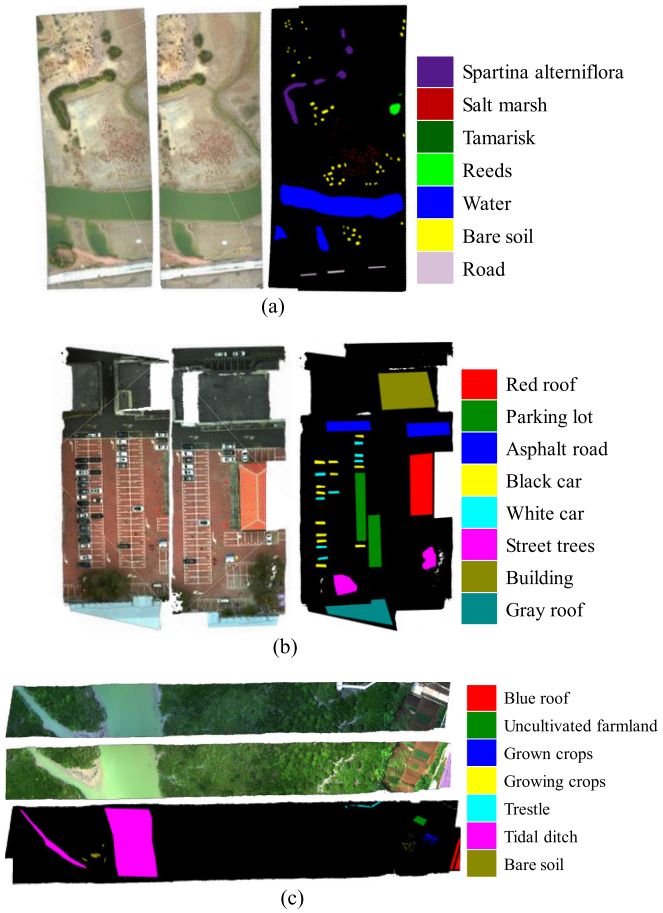


Fig. 4. False color maps and ground reference maps for three datasets. (a) YR. (b) Campus of Harbin Institute of Technology. (c) Shankou Mangrove.

exhibit certain overlap with an overlapping ratio of around 60%, will be used for the experiment. The YR Delta is home to several species of vegetation, including salt marsh, bare soil, reeds, suaeda salsa, tamarix, and so on. Fig. 4(a) depicts original UAV strips and ground reference map.

2) Campus of Harbin Institute of Technology (HIT): This UAV HS dataset was captured at HIT, Harbin, China, in 2021. It was acquired using the aforementioned data acquisition system at a flying height of 80 m, and it has 273 bands at approximately 0.1-m ground resolution and an overlapping ratio of 34%. As it was observed in an urban college setting, the dataset primarily contains categories typically found in urban scenes, such as buildings, vehicles, parking lots, and street trees. Fig. 4(b) shows original UAV strips and ground reference map of this dataset.

3) Shankou Mangrove (Shankou): This UAV HS dataset was acquired in Shankou, Guangxi province of China, in 2021. It was also acquired by the aforementioned data acquisition system, with a flying height of 80 m, a flying speed of 7 m/s, and 273 bands at around 0.1-m ground resolution. Twelve strips were captured over several days, and only two of them, with a partially overlapped region and overlapping ratio of about 35%, will be used in this article. The scene includes various uneven objects, such as grown and growing crops, trestles, and tidal ditches. Fig. 4(c) shows original UAV strips and ground reference map.

In this article, the IID model for UAV image strips necessitates precise registration of input HS images. Therefore, all three datasets are preprocessed by performing cross-correlation matching utilizing selected noise-free and high-quality spectral bands of image strips.

B. Experimental Results and Analysis

1) Experimental Setting: This article introduces a novel IID method capable of achieving reflectance for UAV image scene. To facilitate fair comparison in experiments, the proposed method is compared with four state-of-the-art UAV stitching methods and two-step UAV images stitching plus IID methods. The detailed comparative methods are described as follows.

1) *Raw data:* Original HS UAV image strips are directly stitched after cross-correlation matching registration.
2) *ELA:* An image stitching method adopts multiple local transformations to locally align images.
3) *mTopKRP:* An image stitching method using mTopKRP algorithm to perform warping and blending.
4) *SP:* An image stitching method based on mesh-based warp and total energy function emphasizing different characteristics of warp.
5) *Raw + HIID:* Perform IID operation on directly stitched HS images to produce its reflectance component.
6) *ELA + HIID:* Perform IID operation on the stitched result produced by ELA method to obtain the reflectance component.
7) *mTopKRP + HIID:* Perform IID operation on the stitched result produced by mTopKRP method to obtain the reflectance component.
8) *SP + HIID:* Perform IID operation on the stitched result of SP method to produce its reflectance product.
9) *AutoStitch:* It works from unordered collections of images and automatically finds matches between images using the SIFT algorithm [52].
10) *Proposed:* A targeted HS IID method aiming at stitching two UAV strips.

2) Visual Comparison: Besides quantitative analysis, it is essential to incorporate visual evaluation methods for comparing these methods.

Figs. 5–7 show the stitching results obtained by comparative methods on the three datasets. Observing the stitching outcomes across the three datasets, it is apparent that while

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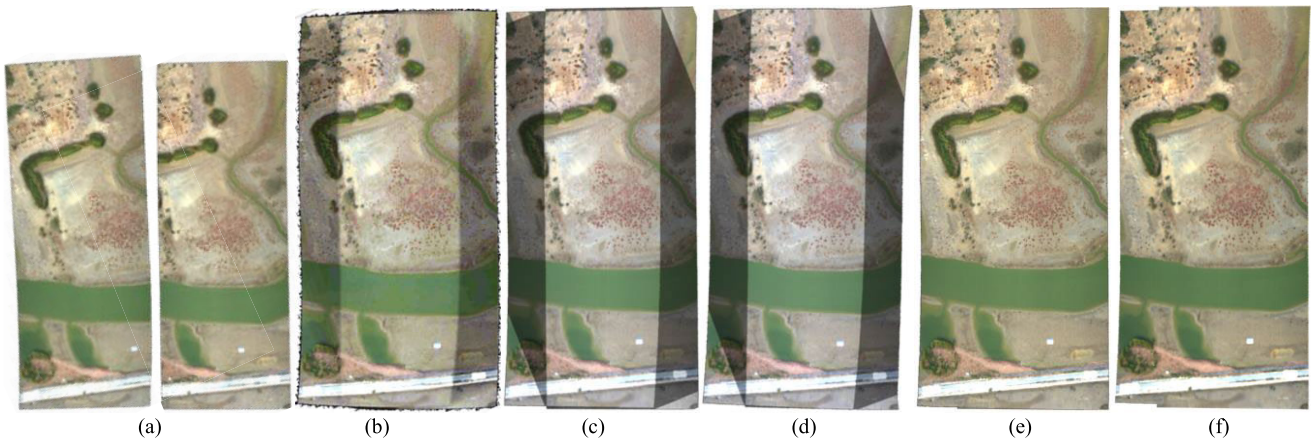


Fig. 5. Stitching results of YR dataset. (a) Source images. (b) AutoStitch. (c) ELA. (d) mTopKRP. (e) SP. (f) Proposed.

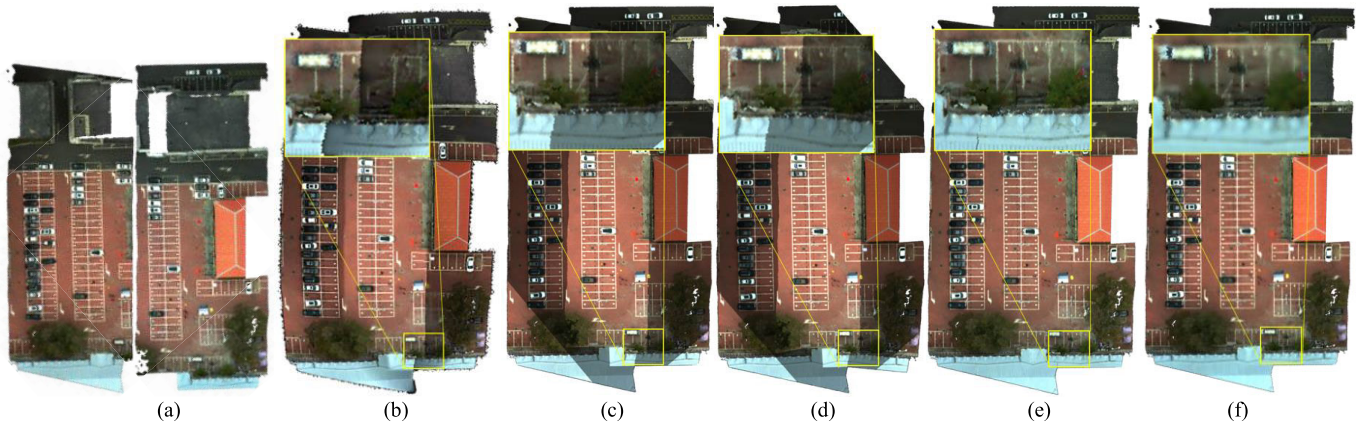


Fig. 6. Stitching results of HIT dataset. (a) Source images. (b) AutoStitch. (c) ELA. (d) mTopKRP. (e) SP. (f) Proposed.

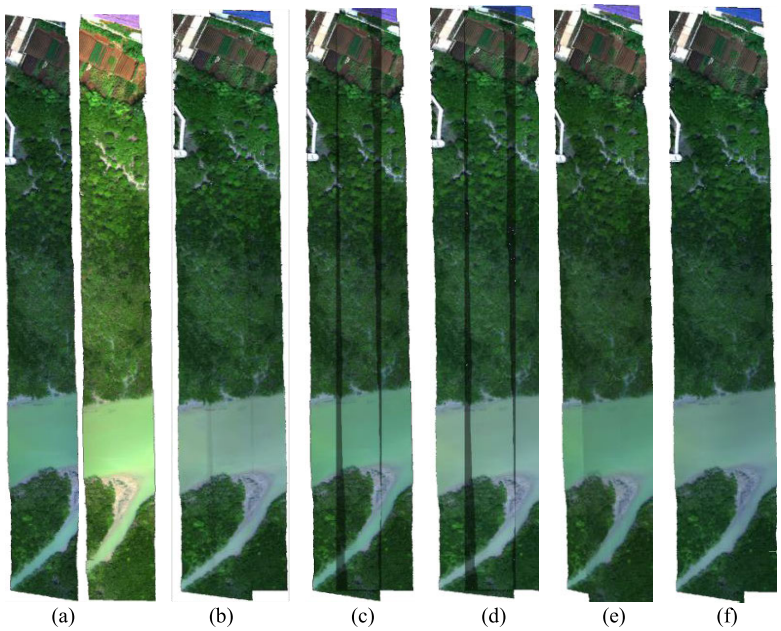


Fig. 7. Stitching results of Shankou dataset. (a) Source images. (b) AutoStitch. (c) ELA. (d) mTopKRP. (e) SP. (f) Proposed.

the AutoStitch method successfully stitches UAV image strips, discernible seam traces persist. With regard to ELA and mTopKRP methods, their stitching traces are more easily discernible, characterized by alternating bright and dark regions across the entire scene. In Fig. 6, for instance, in HIT dataset stitching results, the image produced after stitching

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TABLE I
WCRV RESULTS OF DIFFERENT METHODS ON THREE DATASETS

Data sets	Categories	Raw	Raw-HIID	ELA	ELA-HIID	mTopKRP	mTopKRP-HIID	SP	SP-HIID	Proposed
YR	Spartina alterniflora	1.3968	0.5261	1.2849	1.2979	1.3974	0.5640	2.0169	0.7436	**0.3926**
	Salt marsh	0.2352	0.0511	0.0169	0.0078	0.0085	0.0331	0.0371	0.0056	**0.0029**
	Tamarisk	0.1735	0.0396	0.2000	0.0528	0.1660	0.0433	0.4283	0.0849	**0.0226**
	Reeds	0.0397	0.0058	0.5650	0.4113	0.0035	**0.0006**	0.0402	0.0052	0.0059
	Water	0.0123	0.0026	0.0834	0.0475	0.0851	0.0180	0.0135	0.0027	**0.0019**
	Bare soil	0.6138	0.1722	0.6573	0.5798	0.1894	0.1808	0.1178	0.0823	**0.0331**
	Road	1.9587	0.8705	0.0045	**0.0012**	2.1056	0.8978	4.2106	1.8041	0.9225
	Average	0.6329	0.2383	0.4017	0.3426	0.5651	0.2482	0.9806	0.3898	**0.1974**
HIT	Red roof	0.1456	0.0380	0.0290	0.0083	0.0254	0.0380	0.1458	0.0377	**0.0065**
	Parking lot	0.9757	0.1741	0.7356	0.1821	0.5372	0.1212	0.7039	0.1142	**0.1026**
	Asphalt road	0.1503	0.0311	0.1561	0.0441	0.1425	**0.0276**	0.1452	0.0288	0.0344
	Black car	0.7843	0.1792	0.3658	**0.1074**	0.5117	0.1441	0.8385	0.1726	0.1719
	White car	13.6855	4.8288	7.9461	**3.8816**	9.8534	3.8981	14.1569	4.6173	4.1451
	Street trees	0.3790	0.2139	0.6337	0.5438	0.5575	0.5126	0.3327	0.0806	**0.0601**
	Building	0.0441	0.0087	0.1183	0.0336	0.0987	**0.0081**	0.0469	0.0091	0.0255
	Gray roof	0.1587	0.0393	0.7038	0.2515	0.6762	**0.0361**	0.1822	0.0456	0.0404
	Average	2.0404	0.6892	1.3360	0.6316	1.5503	0.5982	2.0690	0.6382	**0.5733**
Shankou	Blue roof	0.1278	0.0156	0.6616	0.2099	0.4535	0.1323	0.0840	**0.0094**	0.0156
	Uncultivated farmland	0.0455	0.0086	0.0354	0.0073	0.0411	0.0075	0.0392	**0.0057**	0.0072
	Grown crops	0.3051	0.1249	0.2200	0.1379	0.3760	0.1463	0.7465	0.4473	**0.0846**
	Growing crops	0.4147	0.0500	0.2372	0.0457	0.3520	0.0729	0.1473	**0.0400**	0.0629
	Trestle	1.5409	0.7101	1.2150	0.6185	0.2102	**0.0685**	0.1824	0.1312	0.5531
	Tidal ditch	0.0078	0.0034	0.0290	0.0072	0.0280	0.0039	0.0042	0.0031	**0.0009**
	Bare soil	0.0256	**0.0039**	0.0855	0.0199	0.0207	0.0057	0.0182	0.0121	0.0041
	Average	0.3525	0.1310	0.3549	0.1495	0.2117	0.0625	0.1746	0.0927	**0.1041**

with mTopKRP method loses a part of the upper-right region. For a more detailed comparison, we focus on the bottom-right corner region and highlight it with a yellow rectangle. Upon comparative observation, it is evident that the AutoStitch, ELA, and mTopKRP methods all exhibit spectral discontinuities at the seams, with the mTopKRP method causing a certain degree of image distortion. Although the SP method shows fewer noticeable seam artifacts, they are still present. In contrast, the proposed method effectively avoids this issue and maintains good within-class spectral consistency. While the SP method seems to demonstrate satisfactory performance on YR and HIT datasets, seamline issues still persist when applied to the Shankou dataset, as a distinct seamline can be observed throughout the entire image. The proposed method demonstrates superior performance in visual stitching results on these three datasets, without any problems of seam lines or darkness. In summary, our qualitative assessment of these stitching methods concludes that the proposed method performs the best, followed by the SP method, then the AutoStitch method, and finally the ELA and mTopKRP methods.

3) Spectral Consistency Comparison: To evaluate the spectral consistency of comparative methods, the within-class reflectance variance (WCRV) metric is utilized to assess the quality of produced reflectance. It can be expressed in the following manner:

$$\mathrm{WCRV}(\mathbf{R}) = \frac{1}{k}\sum_{i=1}^{k}\sum_{j\in C_i}\left\|\mathbf{R}_j - \mathbf{m}_i\right\|_2^2 \tag{28}$$

where k is the total number of categories, and $\mathbf{m}_i$ is the mean vector of category C_i in reflectance.

Table I illustrates the total WCRV metric for each category on three datasets, along with the average performance of WCRV with each comparative method. When comparing the raw stitched results to images processed only using stitching methods, there is a relatively small change in WCRV metric. Due to variations in datasets, some metrics may increase while others may decrease. However, once the single image intrinsic decomposition is performed after stitching, the produced images demonstrate a significant reduction in WCRV metric compared to results without intrinsic decomposition procedure. Furthermore, this conclusion is confirmed on all three datasets, highlighting the key role of intrinsic decomposition in enhancing within-class spectral consistency. The proposed method achieves not only image stitching but also intrinsic decomposition goals for recovering the reflectance component of entire scene. In addition, based on the results of WCRV, it is evident that the proposed method exhibits superior within-class spectral consistency for the majority of categories and performs optimally in terms of average WCRV performance on three datasets. More specifically, the proposed method shows an average increase of approximately 66.2% and 16.7% for stitching and stitching-IID methods, respectively, indicating the superiority of the proposed method in improving within-class spectral consistency.

In addition, to evaluate the spectral consistency performance in overlapping and nonoverlapping areas of comparative methods, areas from different image strips are directly stitched as a new single image. In this manner, overlapping area of $\mathbf{I}_1$ and nonoverlapping area of $\mathbf{I}_2$ form the image $\mathbf{I}_{12t}$, and similarly, overlapping area of $\mathbf{I}_2$ and nonoverlapping area of $\mathbf{I}_1$ form the image $\mathbf{I}_{21t}$. The spectral standard deviation of homogeneous land cover samples is measured to evaluate the

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TABLE II
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON YR DATASET

Categories	Raw	Raw-HIID	ELA	ELA-HIID	mTopKRP	mTopKRP-HIID	SP	SP-HIID	Proposed
Spartina alterniflora	83.29	84.72	57.95	61.18	58.18	63.33	80.16	80.73	**88.78**
Salt marsh	80.03	80.18	97.14	97.38	99.16	99.66	98.49	**99.89**	85.52
Tamarisk	40.23	47.91	95.30	**95.60**	93.77	94.43	56.97	40.40	63.72
Reeds	84.43	91.38	74.34	70.26	99.72	**100.00**	85.83	83.17	90.91
Water	**100.00**	**100.00**	**100.00**	**100.00**	97.66	95.49	**100.00**	**100.00**	**100.00**
Bare soil	83.19	83.18	80.82	79.44	73.66	73.21	98.76	99.04	**99.49**
Road	99.61	99.58	99.83	**100.00**	78.63	75.98	78.22	83.84	83.17
OA	95.78	96.14	93.21	93.52	91.34	90.26	96.37	96.52	**96.72**
AA	81.54	83.85	86.48	86.27	85.83	86.02	85.49	83.87	**87.37**
Kappa	0.9047	0.9130	0.8144	0.8227	0.7713	0.7519	0.8998	0.9041	**0.9259**

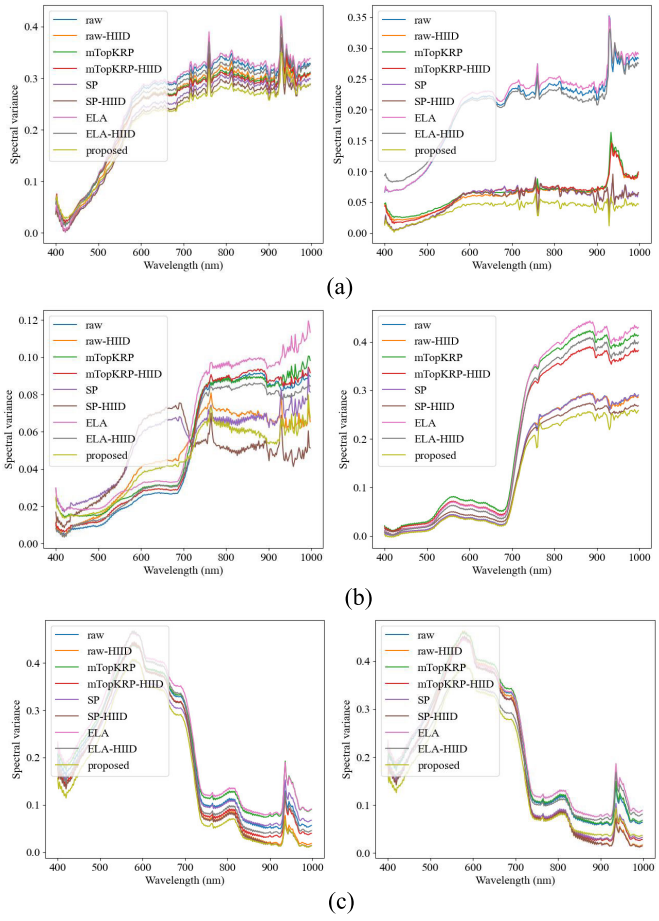


Fig. 8. Spectral standard deviation results of homogeneous land cover on three datasets. (a) Bare soil in YR dataset, (b) street trees in HIT dataset, and (c) tidal ditch in Shankou dataset.

effectiveness of comparative methods in maintaining spectral consistency across different image strips where a lower value signifies higher spectral consistency. The results are depicted in Fig. 8.

Spectral standard deviation in images $\mathbf{I}_{12t}$ and $\mathbf{I}_{21t}$ for YR dataset is presented in Fig. 8(a), where bare soil is the test category due to its samples being distributed across images $\mathbf{I}_{12t}$ and $\mathbf{I}_{21t}$. The results indicate that images produced with stitching and IID processes exhibit lower spectral deviation compared to those without IID process. Moreover, the proposed approach exhibits the lowest spectral deviation among all comparative methods. Similarly, the same conclusion can also be inferred from the experimental results of the remaining two datasets. In Fig. 8(b) and (c), street trees and tidal ditch are used as the testing categories for HIT and Shankou datasets, respectively. IID approach has a significant effect on reducing spectral standard deviation, especially in the wavelength range from 750 to 1000 nm. Furthermore, the proposed method outperforms other stitching and stitching-IID methods on all experimental datasets. Moreover, it is evident that ELA and mTopKRP methods, as well as their IID results, tend to show large spectral deviation on samples scattered over the entire image, indicating that these stitching results suffer from the spectral discontinuity problem of ground surface at the image seam area.

4) Classification Results and Analysis: To evaluate the performance of generated stitched images applied to subsequent tasks, especially classification purpose, all produced results are trained and tested by a k-nearest neighbor (KNN) classifier. The evaluation of spectral information accuracy commonly employs metrics such as overall accuracy (OA), Kappa, and average accuracy (AA). Ten training samples from each category are randomly selected to train the classifier, and reported results represent the average value of ten experimental runs on randomly selected training and testing samples. Figs. 9–11 illustrate the classification results of comparative methods on three UAV datasets. It can be seen that while both ELA and mTopKRP methods can successfully accomplish the stitching task, neither of them produces images with good spectral consistency between UAV strips after stitching process on three datasets. The same problem is also observed in ELA and mTopKRP stitching results following the implementation of IID. Regarding the raw datasets, the directly stitched results exhibit no anomalies because of the unobvious spectral variations between the two strips in YR and HIT datasets. However, as depicted in Fig. 11, the classification map of directly stitched image exhibits a sudden change at the seam location in Shankou dataset. Similarly, the SP stitching method also has the same problem, as evidenced by abrupt change at the seams of the stitched classification images found in Fig. 11 as well. By comparison, SP performs better stitching than ELA and mTopKRP methods in terms of spectral quality. In addition, Figs. 9–11 depict the classification results in three datasets produced by the proposed method, from which it can be observed that the proposed method yields high-quality

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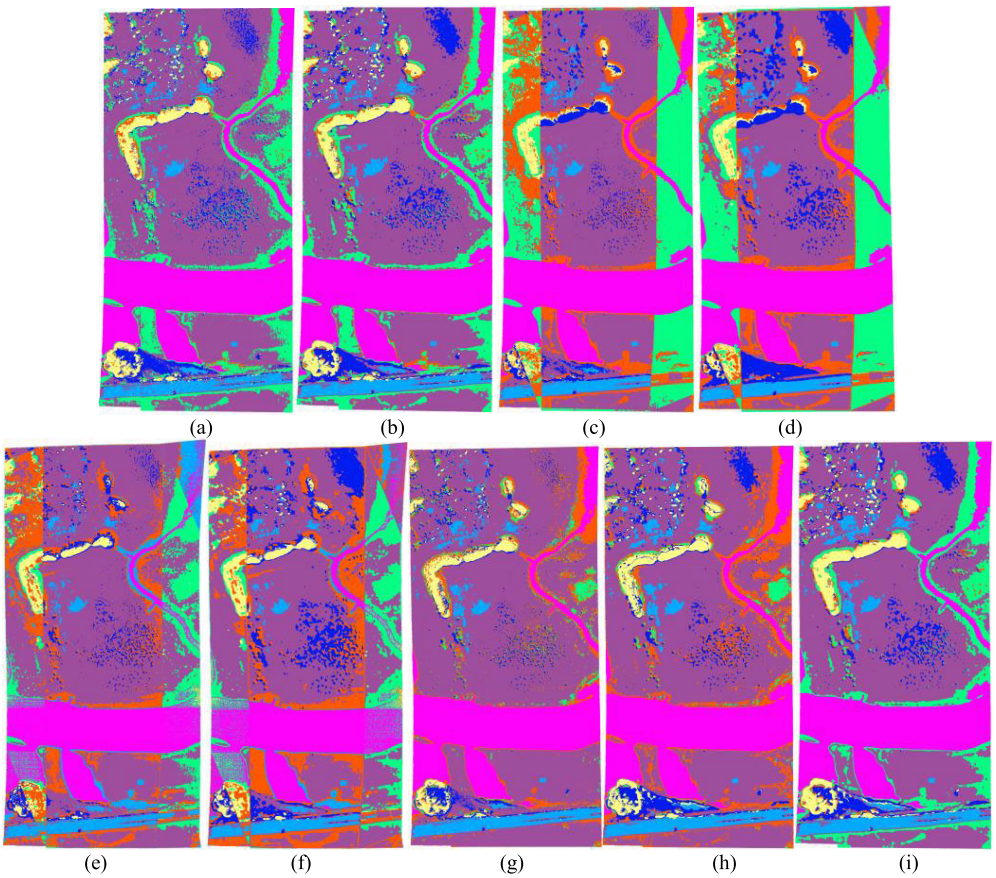


Fig. 9. Classification results of YR dataset. (a) Raw. (b) Raw-IID. (c) ELA. (d) ELA-IID. (e) mTopKRP. (f) mTopKRP-IID. (g) SP. (h) SP-IID. (i) Proposed.

TABLE III
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON HIT DATASET

Categories	Raw	Raw-HIID	ELA	ELA-HIID	mTopKRP	mTopKRP-HIID	SP	SP-HIID	Proposed
Red roof	93.34	92.72	96.32	96.43	**96.69**	96.57	93.71	92.64	92.46
Parking lot	89.91	96.86	92.70	98.02	95.18	**98.48**	91.20	98.18	97.94
Asphalt road	90.89	91.32	**93.22**	90.16	92.06	90.64	92.03	90.54	90.56
Black car	75.01	**75.63**	67.45	67.20	63.08	64.16	73.94	74.56	75.52
White car	51.55	52.69	36.56	38.42	39.85	40.46	53.40	**53.52**	50.69
Street trees	97.83	97.89	97.94	98.42	98.92	99.25	99.15	99.40	**99.68**
Building	90.52	91.45	87.02	86.68	85.31	85.78	89.41	91.21	**94.89**
Gray roof	99.37	**99.71**	97.79	97.61	97.79	98.01	99.12	99.70	98.02
OA	91.12	92.59	90.72	91.23	90.45	91.06	91.27	92.53	**93.43**
AA	86.05	87.28	83.63	84.12	83.61	84.17	86.49	**87.47**	**87.47**
Kappa	0.8939	0.9114	0.8892	0.8951	0.8865	0.8936	0.8958	0.9107	**0.9213**

TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON SHANKOU DATASET

Categories	Raw	Raw-HIID	ELA	ELA-HIID	mTopKRP	mTopKRP-HIID	SP	SP-HIID	Proposed
Blue roof	93.43	98.99	53.74	54.44	60.61	68.18	99.09	99.19	**99.49**
Uncultivated farmland	99.48	99.81	99.90	**100.00**	99.49	99.80	99.62	99.82	99.39
Grown crops	87.94	88.10	97.04	97.36	93.53	94.28	88.20	89.78	**99.42**
Growing crops	87.52	96.90	90.36	95.30	84.70	92.25	93.36	**97.85**	88.67
Trestle	99.11	99.09	**99.96**	99.41	97.61	97.59	99.26	99.22	98.79
Tidal ditch	99.89	99.89	91.70	92.34	91.38	91.91	99.89	99.89	**100.00**
Bare soil	**100.00**	**100.00**	94.75	94.12	98.80	98.67	**100.00**	**100.00**	**100.00**
OA	95.66	96.31	94.70	94.88	93.70	94.92	96.18	96.86	**98.72**
AA	95.34	97.54	89.64	90.42	89.44	91.81	97.06	97.96	**97.97**
Kappa	0.9446	0.9529	0.9316	0.9339	0.9191	0.9348	0.9513	0.9599	**0.9834**

classification results on all three datasets. In fact, the proposed method processes IID during UAV HS images stitching, thereby avoiding spectral distortion as much as possible and accurately recovering the reflectance component.

To illustrate the improved classification performance of the proposed method on experimental datasets, Tables II–IV summarize the classification results of comparative methods on different datasets. Consistent with the conclusions derived

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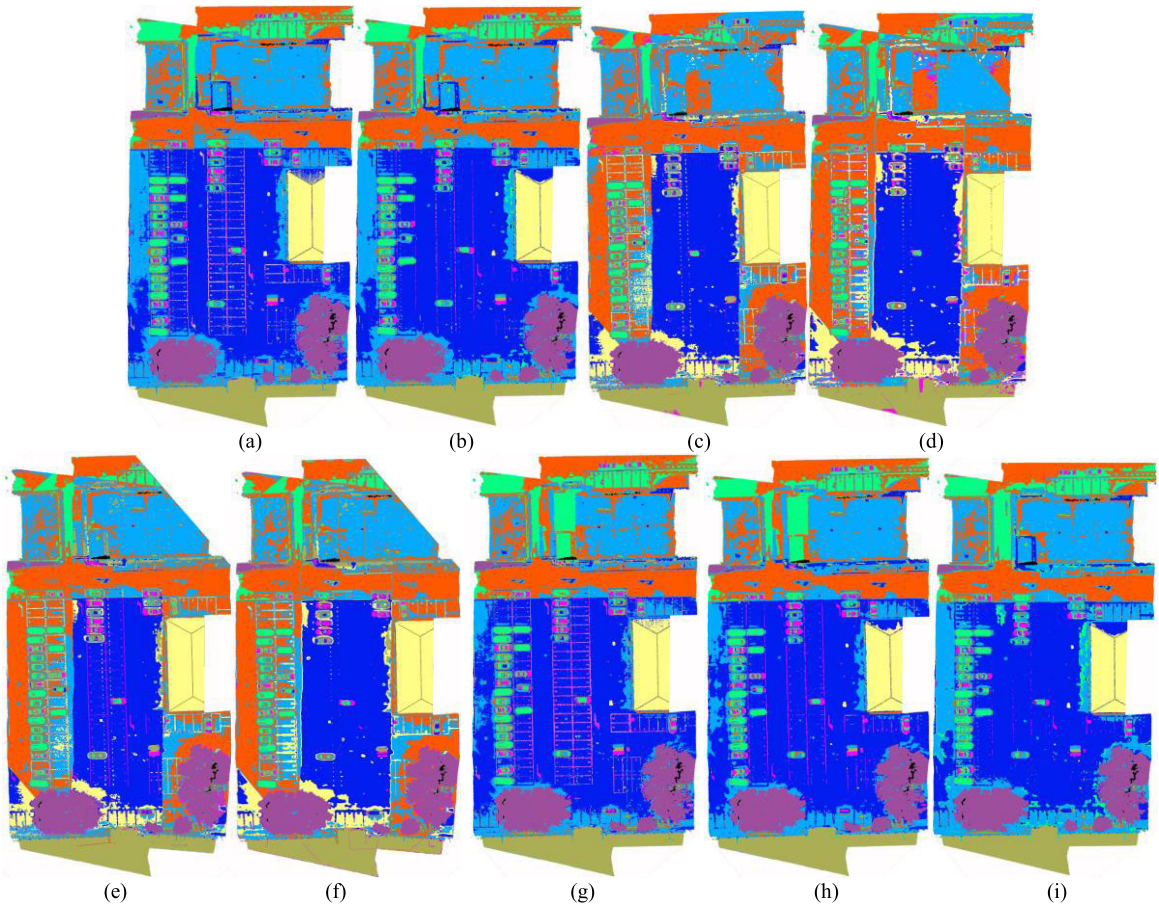


Fig. 10. Classification results of HIT dataset. (a) Raw. (b) Raw-IID. (c) ELA. (d) ELA-IID. (e) mTopKRP. (f) mTopKRP-IID. (g) SP. (h) SP-IID. (i) Proposed.

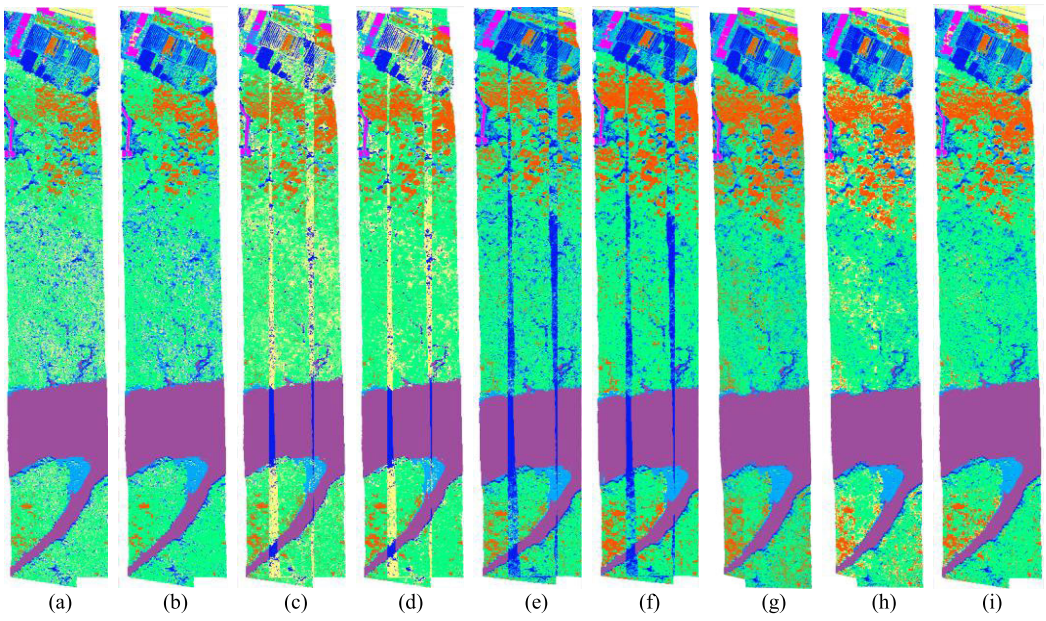


Fig. 11. Classification results of Shankou dataset. (a) Raw. (b) Raw-IID. (c) ELA. (d) ELA-IID. (e) mTopKRP. (f) mTopKRP-IID. (g) SP. (h) SP-IID. (i) Proposed.

in Figs. 9–11, ELA and mTopKRP methods exhibit the lowest classification accuracy due to the distortion of spectral information in certain regions of stitched image scene. However, the directly stitched results and SP methods do not have such problems, and SP method outperforms the other three stitching methods in terms of classification accuracy on all testing datasets. Upon analyzing the classification results, IID technology significantly boosts classification performance.

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When comparing four stitching and four stitching-IID methods in terms of the quantitative evaluation results of classification accuracy, the proposed method demonstrates superiority by attaining the highest OA, AA, and Kappa evaluation. Specifically, the proposed method shows an average increase of 2.9% and 2.4% in OA for three datasets compared to the stitching and stitching-IID methods, respectively, highlighting the robust improvement capability of the proposed method across different datasets.

V. Conclusion

From the perspective of UAV image strips, two major challenges hinder their efficient utilization: 1) spectral uncertainty due to various interferences from natural factors and 2) limited width of UAV image strips impeding subsequent application of large scene. To address these challenges, this article introduces an innovative HS IID model for UAV image strips, which aims to eliminate interference caused by natural factors within and between image strips and accurately recover the reflectance component of overlapping and nonoverlapping regions. The proposed method includes the following steps: initially, an IID energy constraint for UAV strips is introduced to model the IID process of two partially overlapped UAV images. Second, an anti-shadow interference mechanism is integrated to reduce the impact of shadow in overlapping region. Third, a spectral correction constraint is introduced to rectify the spectral bias between overlapping and nonoverlapping regions. Experimental results illustrate that the proposed method effectively removes shading interference and accurately recovers the reflectance component of UAV image strips. However, the proposed approach can only tackle the IID problem of stitching two UAV image strips and requires precise pixel-level registration between two image strips. Therefore, in future studies, we intend to develop an HS IID approach that is more universally applicable, eliminating the need for pixel-level registration, and specifically tailored for stitching UAV strips in large-scale scenes.

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